

Development of a Mock-up of a Smart Insole Using FSR and ADXL345

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Abstract

Background: The research article discusses a mock-up of a Smart Insole prototype using Force Sensitive Resistors (FSRs) and an ADXL345 accelerometer.

Aim & Objective: The primary aim of this research is to create a Smart Insole prototype that integrates advanced sensor technologies to monitor and analyze foot biomechanics. This project focuses on the practical implementation of a mock-up insole that highlights the potential applications in healthcare, sports performance enhancement, and rehabilitation.

Materials & Methods: The paper details the creation and implementation of a "Smart Insole" prototype that incorporates Force Sensitive Resistors (FSRs) and the ADXL345 accelerometer. This prototype is designed with FSRs strategically embedded within the insole to measure pressure distribution across different areas of the foot. Additionally, it utilizes the ADXL345 accelerometer to monitor and record the foot's orientation and movement, providing a comprehensive analysis of foot biomechanics.

Result: The project is focused on developing an innovative wearable technology that monitors foot biomechanics and aims to enhance human performance through detailed analysis and data-driven insights.

Conclusion: By integrating sophisticated hardware, developing firmware with Arduino IDE, and employing the Processing application for visualization, the Smart Insole provides real-time monitoring and analysis of gait dynamics, pressure distribution, and foot movement.

Future Scope: The creation of the Smart Insole mock-up marks a crucial milestone in advancing wearable technology solutions for monitoring foot biomechanics and enhancing human performance. While still in its prototype phase, the mock-up showcases impressive functionality and reveals significant potential for applications in healthcare, sports performance analysis, and rehabilitation.

Keywords: Gait Analysis, Pressure Mapping, Orientation Tracking, Force Sensitive Resistor, Accelerometer

1 Introduction

Assessing foot health is crucial for preventing injuries, improving mobility, and maintaining overall well-being, as the feet are fundamental for daily activities and physical function [1]. This mock-up aims to develop a smart insole prototype utilizing FSR and IMU sensors to monitor real-time feedback on foot pressure distribution, movement, and orientation. By addressing these limitations, we seek to empower individuals with actionable insights into their foot health, enabling timely intervention and improved quality of life [2]. The meticulously planned and executed mock-up underscores the practicality and potential applications of Smart Insole technology in healthcare, rehabilitation, and sports performance measurement [3]. Existing methods for assessing foot health lack accessibility and real-time monitoring capabilities, leading to challenges in proactive management of foot-related conditions [4]. The primary objective of this project is to develop a smart insole prototype integrating FSR and IMU sensors to provide real-time feedback on foot pressure distribution, movement, and orientation, aiming to empower individuals with actionable insights into their foot health for proactive management and prevention of potential complications [5]. Additionally, the project aims to contribute to the advancement of wearable technology by addressing challenges related to foot health monitoring and promoting proactive management of foot-related conditions [4]. By embedding these sensors discreetly into an insole, we envision a wearable device that can revolutionize the way we understand and monitor foot dynamics, offer insights into gait analysis, balance, and foot health [6].

2 Background

The background centers on the growing interest in wearable technology solutions for monitoring and optimizing human performance, particularly in the context of foot biomechanics and gait analysis [7]. Traditional methods of assessing gait dynamics often rely on expensive laboratory equipment and specialized settings, limiting accessibility and scalability [8]. In response to this challenge, there has been a surge in research and development efforts aimed at creating portable, non-invasive solutions capable of providing real-time insights into foot pressure

distribution and movement patterns [9]. Leveraging advancements in sensor technology, the incorporation of Force Sensitive Resistors (FSRs) and the ADXL345 accelerometer within a Smart Insole prototype offers a promising avenue for addressing these needs [10]. By combining sensor data acquisition with data processing and visualization techniques, the Smart Insole holds promise for revolutionizing gait analysis, sports performance assessment, and rehabilitation practices, fostering a new era of personalized and accessible healthcare solutions [11].

3 Materials and Methodology

The project encompasses a broad spectrum of research and developments in wearable technology, foot health monitoring, and sensor technologies. The methodology delves into algorithm development for sensor data analysis, user interface design principles, applications in sports rehabilitation and diagnostics, and consumer acceptance of wearable health monitoring devices.

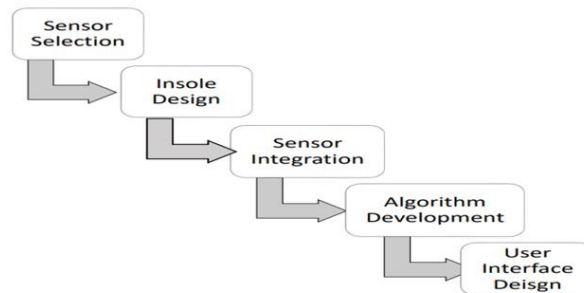


Figure 1. Methodology Followed for the Development of the Smart Insole

3.1 Sensor Selection:

Thoroughly evaluating various Force-Sensitive Resistor (FSR) and Inertial Measurement Unit (IMU) sensors based on technical specifications, accuracy, and suitability for foot health monitoring, ensuring optimal sensor performance. Force Sensitive Resistors (FSRs) are sensors that detect variations in resistance in response to exerted force or pressure. They consist of a conductive polymer material whose resistance decreases as force is applied to its surface. This change in resistance is proportional to the magnitude of the applied force, allowing FSRs to provide continuous feedback on pressure levels. This working principle enables FSRs to be used in various applications such as touch sensing, force measurement, and pressure mapping, including their integration into the Smart Insole prototype for monitoring foot biomechanics and gait analysis [12]. An Inertial Measurement Unit (IMU) is a sensor system that measures and reports accelerations, angular velocities, and sometimes magnetic fields. It typically uses a combination of gyroscopes, magnetometers, and accelerometers. In the Smart Insole project, the ADXL345 accelerometer, an integral component of the IMU, is employed to monitor foot orientation and movement dynamics, contributing to the analysis of gait patterns and foot biomechanics [13].

3.2 Insole Design:

Meticulously crafting the design of the smart insole, considering factors such as material selection, sensor placement, and ergonomic considerations to ensure both comfort and functionality for the user [14].

3.3 Sensor Integration:

Embedding FSR and IMU sensors into the insole, ensuring precise wiring and connections to guarantee reliable data transmission and accurate capture of foot dynamics [15].

3.4 Algorithm Development:

Developing sophisticated algorithms for processing and analyzing sensor data, employing advanced techniques for noise reduction, signal filtering, and feature extraction to derive meaningful insights into foot health metrics [16].

3.5 User Interface Design:

Creating a system for real-time monitoring of foot health data, prioritizing simplicity, usability, and clear visualization of sensor data to enhance user engagement and understanding [17].

4 Development And Execution

The design and planning phase of the Smart Insole mock-up development involved comprehensive consideration of key factors to ensure the effectiveness and functionality of the prototype. The process began with a thorough

assessment of the intended application scenarios, user requirements, and technical specifications. Initial discussions and brainstorming sessions were conducted to define the objectives and scope of the project. This included identifying the target user demographic such as athletes [18], patients undergoing rehabilitation [19], and individuals seeking to improve gait dynamics [20]. Understanding user needs and expectations played a crucial role in informing design decisions and feature prioritization. The physical design of the Smart Insole mock-up involved conceptualizing the layout, form factor, and integration of components. Considerations were made regarding the placement of Force Sensitive Resistors (FSRs), the ADXL345 accelerometer, and the Arduino Uno R3 within the insole base to optimize sensor coverage and data accuracy.

4.1 Analyzing the Gait Cycle

The gait cycle, alternatively referred to as the walking or locomotion cycle, encompasses the sequence of movements involved in human walking. It consists of two main phases: the stance phase and the swing phase. During the stance phase, the foot contacts the ground, beginning with the heel strike as the initial point of contact, followed by the midstance where the foot bears the body's weight, and concluding with the toe-off as the foot pushes off from the ground to initiate the swing phase. The swing phase involves the leg swinging forward to advance the body's position, characterized by the swing-through motion of the foot until the next heel strike occurs [21].

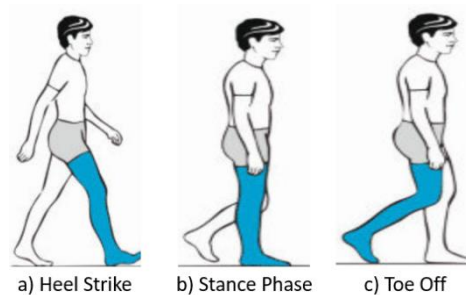


Figure 2. The Three Stages of Gait Cycle Tracked Using the Smart Insole: a) Heel Strike, b) Stance Phase, c) Toe Off

4.2 Integration of FSRs and ADXL345

The incorporation of Force Sensitive Resistors (FSRs) within the Smart Insole prototype is strategically designed to measure and the pressure distribution across key areas of the foot during various phases of the gait cycle [22]. With one FSR positioned at the heel, one at the midfoot, and one at the forefoot (toes), the system aims to provide comprehensive insight into foot biomechanics and pressure dynamics [23].

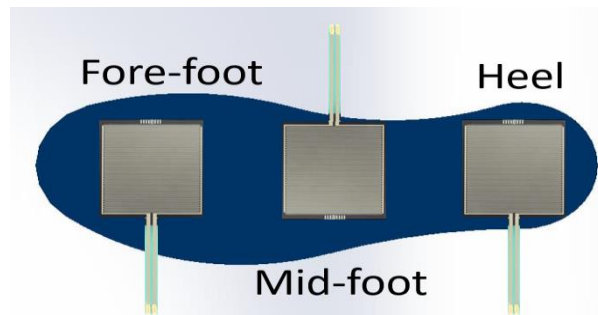


Figure 3. Placement of the 3 FSRs on the Insole

Placing an FSR at the heel allows for the detection of initial ground contact during the heel-strike phase of the gait cycle. This sensor captures the distribution of pressure as weight is transferred from the heel to the midfoot and forefoot during the subsequent phases of stance and propulsion [24]. The FSR at the midfoot provides additional data on pressure distribution as the foot rolls forward and redistributes weight from the heel to the forefoot. Finally, positioning an FSR at the forefoot, near the toes, enables the measurement of pressure exerted during toe-off, the final phase of the gait cycle where the foot pushes off from the ground [25]. By integrating FSRs at these strategic locations, the Smart Insole prototype can effectively monitor pressure variations across the entire foot, offering valuable insights into gait dynamics, foot loading patterns, and potential areas of imbalance or discomfort [26]. This comprehensive pressure mapping capability enhances the utility of the Smart Insole for applications in healthcare, sports performance analysis, and rehabilitation, enabling personalized

monitoring and optimization of foot biomechanics [27]. The strategic placement of the ADXL345 accelerometer near the heel of the Smart Insole is intended to record important information about the dynamics of movement and foot orientation [28]. When the ADXL345 is positioned at the heel, it can monitor the first contact phase of the gait cycle, which is when the heel hits the ground and exerts large forces. Because of its positioning, the sensor can identify changes in the direction and intensity of heel strikes, which offers important information about possible imbalances and gait patterns [15]. The ADXL345 may further add to a thorough examination of foot motion by monitoring acceleration along the x, y, and z axes. This allows for the collection of information on heel lift and the change from the propulsion phase to the midstance phase [29].

4.3 Circuit Development:

The development of the circuit for integrating three Force Sensitive Resistors (FSRs), three 10k ohm resistors, and an Arduino Uno R3 microcontroller unit (MCU) is a pivotal step in creating the Smart Insole prototype for monitoring foot biomechanics. The circuit design aims to accurately measure pressure distribution across the foot and facilitate data acquisition for subsequent analysis. The circuit configuration employs a voltage divider setup for each FSR, comprising the FSR and a 10k ohm resistor connected in series between the supply voltage (VCC) and ground. The voltage output from the junction between the FSR and resistor is proportional to the resistance of the FSR, which varies with applied pressure. This varying voltage is then fed into the analog input pins of the Arduino Uno R3 for analog-to-digital conversion. Through this arrangement, each FSR's pressure-dependent resistance is converted into a corresponding analog voltage signal, which the Arduino Uno R3 reads and processes. The MCU interprets these voltage signals and translates them into meaningful pressure readings, allowing for real-time monitoring of foot pressure distribution.

5 Integration With Arduino IDE

The Arduino IDE serves as the development platform for programming the Arduino Uno R3 microcontroller unit (MCU) embedded within the Smart Insole prototype. Custom firmware is developed using the Arduino programming language, facilitating the configuration of sensor interfaces, data processing algorithms, and communication protocols.

5.1 Establishing FSR Connectivity Using Arduino IDE

Integrating Force Sensitive Resistors (FSRs) with Arduino IDE was a crucial step in the development of the Smart Insole prototype, facilitating real-time monitoring of foot pressure distribution. The process began with configuring the Arduino Uno R3 microcontroller unit (MCU) to interface with the FSRs and acquire analog sensor data. In the Arduino IDE, a dedicated sketch is developed to initialize analog input pins corresponding to the FSR connections. Each FSR is connected to a specific analog input pin, along with a 10k ohm resistor forming a voltage divider circuit. This configuration ensures accurate measurement of voltage variations across the FSRs in response to applied pressure. These calibrated pressure readings are then processed and transmitted for visualization and analysis, laying the foundation for comprehensive monitoring of foot biomechanics using the Smart Insole prototype.

5.2 Linking ADXL345 Via Arduino IDE

In the Arduino IDE, a dedicated sketch is developed to initialize the communication interface and configure the necessary registers within the ADXL345 sensor module. This includes setting up parameters such as data rate, measurement range, and power modes to optimize sensor performance for the specific application requirements. The Arduino IDE sketch further includes routines for reading accelerometer data from the ADXL345 sensor module, parsing the raw sensor readings, and converting them into meaningful units such as acceleration along three axes (x, y, z). These calibrated acceleration values are then processed and utilized for monitoring foot orientation, detecting movement patterns, and analyzing gait dynamics within the Smart Insole prototype. Through the seamless integration of ADXL345 connectivity within the Arduino IDE environment, the Smart Insole prototype becomes a powerful tool for assessing foot biomechanics, enhancing sports performance analysis, and facilitating rehabilitation practices. This linkage between the ADXL345 accelerometer and Arduino IDE forms the backbone of the Smart Insole's sensor system, enabling comprehensive monitoring and analysis of human movement.

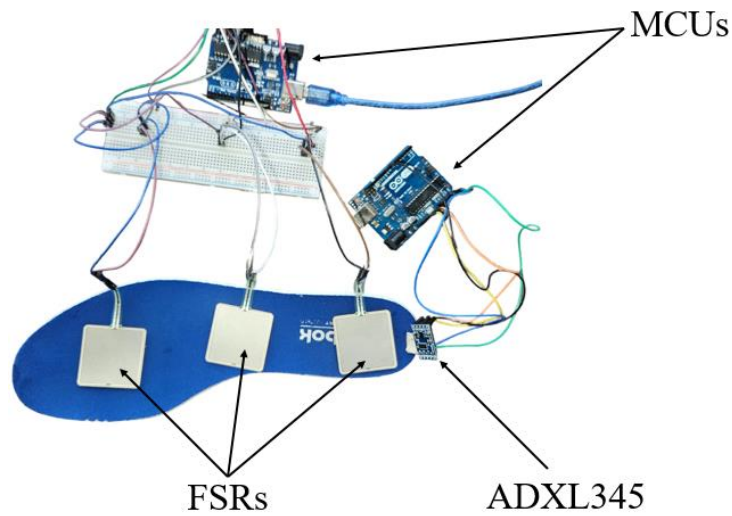


Figure 4. Prototype of Smart Insole Utilizing FSR and ADXL345

6 Results and Discussion

The integration of Force Sensitive Resistors (FSRs) and the ADXL345 accelerometer with the Arduino Uno R3 microcontroller unit (MCU) yielded real-time data output on the Arduino Serial Monitor. The FSRs provided continuous pressure readings, reflecting changes in foot pressure distribution during different phases of the gait cycle. Concurrently, the ADXL345 accelerometer delivered acceleration values along three axes, offering insights into foot orientation and movement dynamics. This real-time feedback on foot biomechanics enables users to monitor gait patterns, detect abnormalities, and optimize performance. The seamless integration of FSR and ADXL345 connectivity with the Arduino IDE environment demonstrates the feasibility and effectiveness of the Smart Insole prototype for future comprehensive monitoring and analysis of foot biomechanics. Upon integration with the Arduino Uno R3 microcontroller unit (MCU) and configuration within the Arduino IDE, the FSRs present real-time data output on the Arduino Serial Monitor. As pressure is applied to the FSRs embedded within the Smart Insole prototype, corresponding analog voltage values are captured and transmitted via the analog input pins of the Arduino Uno R3. These voltage values are converted into digital readings using the built-in analog-to-digital converter (ADC) of the MCU.

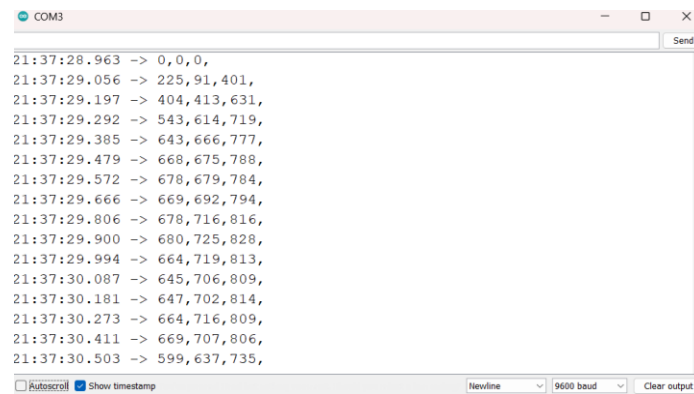


Figure 5. Serial Monitor Readings of the FSRs

On the Arduino Serial Monitor, users can observe the continuous stream of pressure readings from each FSR, providing insights into foot pressure distribution throughout different phases of the gait cycle. Through this visualization, users can monitor changes in pressure patterns, detect anomalies, and assess foot biomechanics in real time, enhancing the utility of the Smart Insole for gait analysis and performance optimization. The integration of the ADXL345 accelerometer with the Arduino Uno R3 MCU enables real-time data output on the Arduino Serial Monitor. The accelerometer continuously measures acceleration along three axes (x, y, z) and transmits raw sensor readings to the MCU. On the Arduino Serial Monitor, users can observe the real-time acceleration values corresponding to each axis, providing insights into foot orientation and movement dynamics. Changes in readings reflect variations in foot position during the gait cycle. By visualizing these data on the Arduino Serial Monitor, users can monitor foot movement patterns, detect deviations from normal gait, and assess balance and stability in

real time.

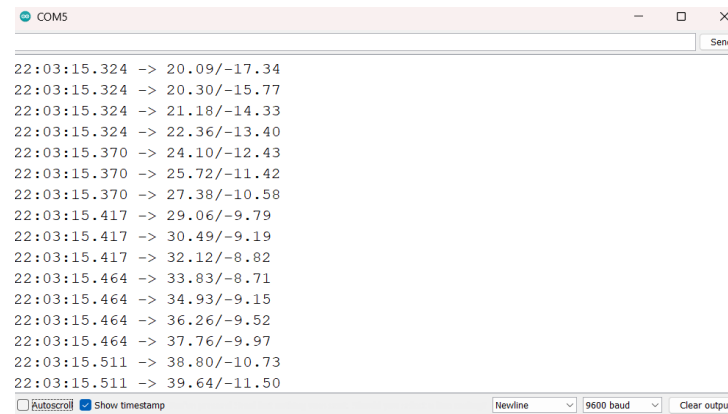


Figure 6. Serial Monitor Readings of the ADXL345

Through the seamless integration of ADXL345 connectivity within the Arduino IDE environment, the Smart Insole prototype becomes a powerful tool for assessing foot biomechanics, enhancing sports performance analysis, and facilitating rehabilitation practices. This linkage between the ADXL345 accelerometer and Arduino IDE forms the backbone of the Smart Insole's sensor system, enabling comprehensive monitoring and analysis of human movement.

7 Result Visualization Using Processing Application

The integration of the Smart Insole prototype with the Processing application facilitated real-time visualization of sensor data, enhancing the interpretability and usability of the system. The Processing application received sensor data streams from the Arduino Uno R3 microcontroller unit (MCU) via serial communication, allowing for dynamic visualization of foot pressure distribution and movement dynamics.

7.1 Visualizing FSR Readings Via Processing

For Force Sensitive Resistors (FSRs), the Processing application generated visual pressure profiles, providing intuitive representations of pressure distribution across the foot. Regions of elevated pressure, such as the heel or forefoot, were depicted with distinct colors upon changing intensity, enabling users to identify pressure hotspots and assess foot-loading patterns throughout various phases of the gait cycle. The visualization was obtained for different gait patterns such as no pressure moment, heel strike, stance phase, toe off, flat feet.

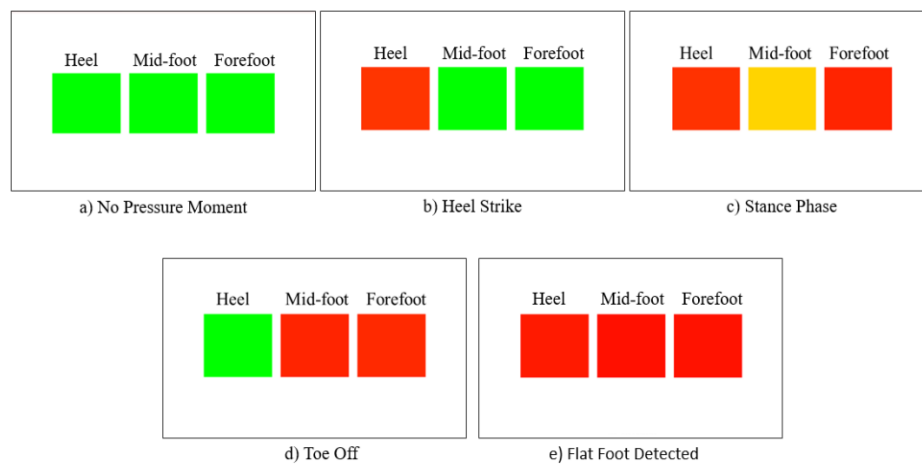


Figure 7. Visualization Obtained from FSRs of Various Stages of a Gait Cycle: a.) No Pressure, b.) Heel Strike, c.) Stance Phase, d.) Toe Off, e.) Flat Feet Detection

7.2 Visualizing FSR Readings Via Processing:

For the ADXL345 accelerometer, the Processing application visualized foot orientation and movement dynamics using graphical representations of 3D animations. Changes in acceleration along different axes were depicted in real time, allowing users to monitor foot movement patterns, detect deviations from normal gait, and assess balance and stability.

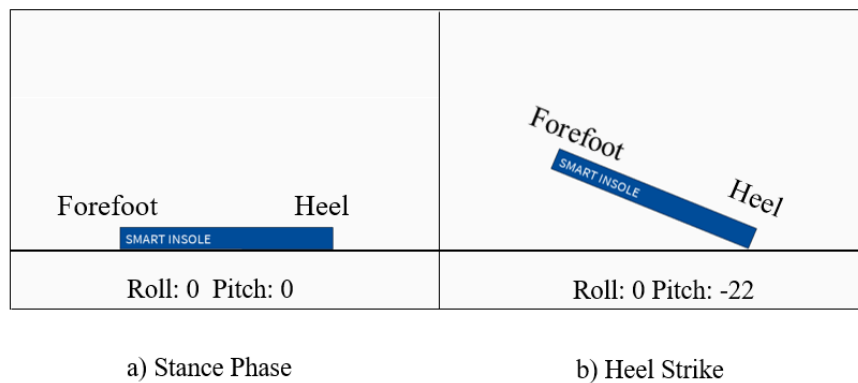


Figure 8. Visualizations of the Orientation of the Insole Obtained from the ADXL345 at: a) Stance Phase, b) Heel Strike

8 Conclusion and Future Scope:

The development of the Smart Insole mock-up represents a significant milestone in the journey towards realizing wearable technology solutions to track foot biomechanics and improve human performance. Despite its status as a prototype, the mock-up demonstrates promising functionality and potential applications in healthcare, sports performance analysis, and rehabilitation. Through meticulous design, integration, and testing, the Smart Insole mock-up successfully showcases the feasibility of incorporating Force Sensitive Resistors (FSRs), an ADXL345 accelerometer, and an Arduino Uno R3 microcontroller unit (MCU) into a wearable form factor. While accurate testing was not feasible at this stage, preliminary assessments confirmed the proper functioning of the system components at a small scale, laying the groundwork for further development and refinement.

Moving forward, comprehensive testing and evaluation of the Smart Insole prototype will be pursued to validate its performance, accuracy, and reliability under diverse conditions. This will involve collaboration with domain experts, healthcare professionals, and end-users to ensure that the prototype meets the needs and expectations of its intended users.

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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